LLaMA Factory

- **LLaMA Factory**는 대형 언어 모델(LLM)을 **쉽고 빠르게 파인튜닝(Fine-tuning)** 할 수 있게 도와주는 오픈소스 프레임워크입니다.
- "코드 거의 없이, GUI 또는 간단한 명령어만으로 LLM을 내 데이터로 학습시키는 도구"

왜 쓰는가?

기존 방법(Hugging Face 등)	LLaMA Factory
복잡한 코드 작성 필요	설정 파일 또는 WebUI로 해결
각 모델마다 다른 설정	100+ 모델을 통합 지원
GPU 메모리 최적화 어려움	QLoRA 등 자동 지원

주요 특징

- **모델**: LLaMA, Mistral, Qwen, Gemma, Phi, ChatGLM 등 100개 이상
- **학습 방식**:
  - Full Fine-tuning (전체 학습)
  - **LoRA / QLoRA / DoRA**
  - RLHF (보상 기반 강화학습)
  - DPO, PPO, ORPO 등 최신 기법
- **인터페이스**:
  - WebUI (클릭으로 조작)
  - CLI (명령어)
  - Python API

WebUI

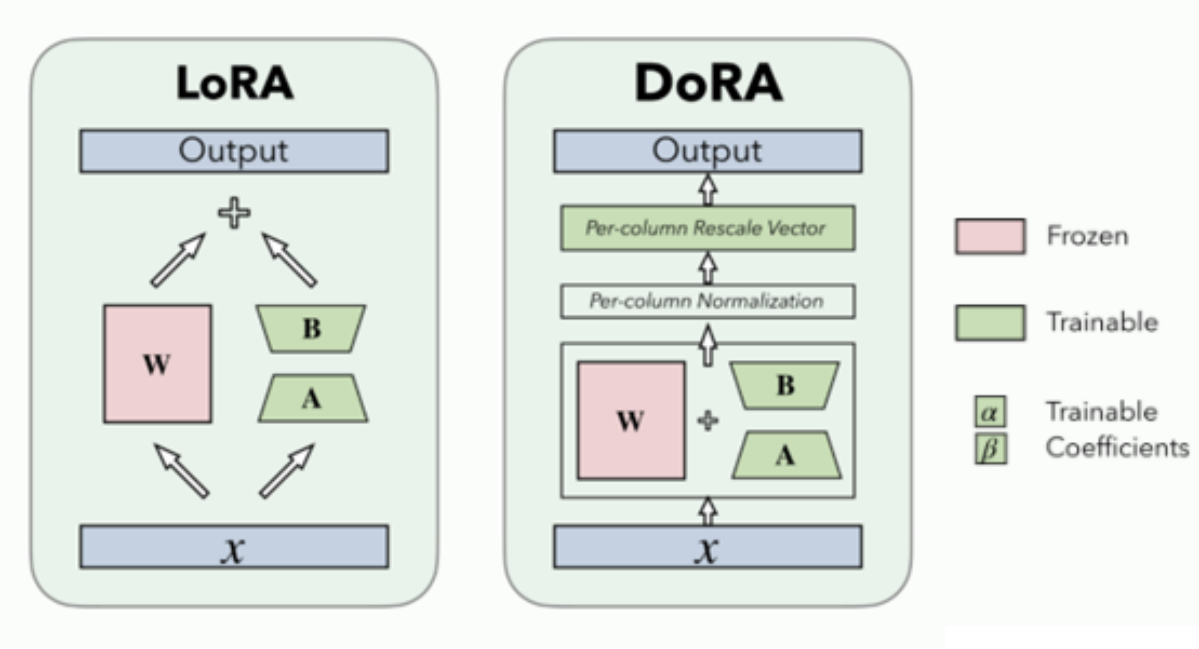
DoRA (Weight-Decomposed LoRA)란?

- DoRA는 모델의 가중치를 **크기(Magnitude)** + **방향(Direction)** 으로 분해하여 학습하는 방식
- **장점**:
  - Full Fine-tuning과 유사한 성능
  - 더 적은 GPU 메모리 사용

DoRA 작동 방식

> DoRA는 LoRA에 두 가지 트릭을 추가한 구조

1. **Per-column Normalization**: 행렬의 각 열(column)을 **개별적으로 정규화**
2. **Per-column Rescale Vector**: 각 열에 대한 스케일링 벡터(m)를 별도 학습



LoRA vs DoRA

항목	LoRA	DoRA
방식	저랭크 행렬 추가	가중치를 크기(magnitude) + 방향(direction)으로 분해
성능	효율적이지만 Full FT와 차이 있음	Full Fine-tuning에 더 근접
GUI 설정	Fine-tuning Method: LoRA	LoRA 선택 후 **"Use DoRA" 체크**

DoRA 설정 예시

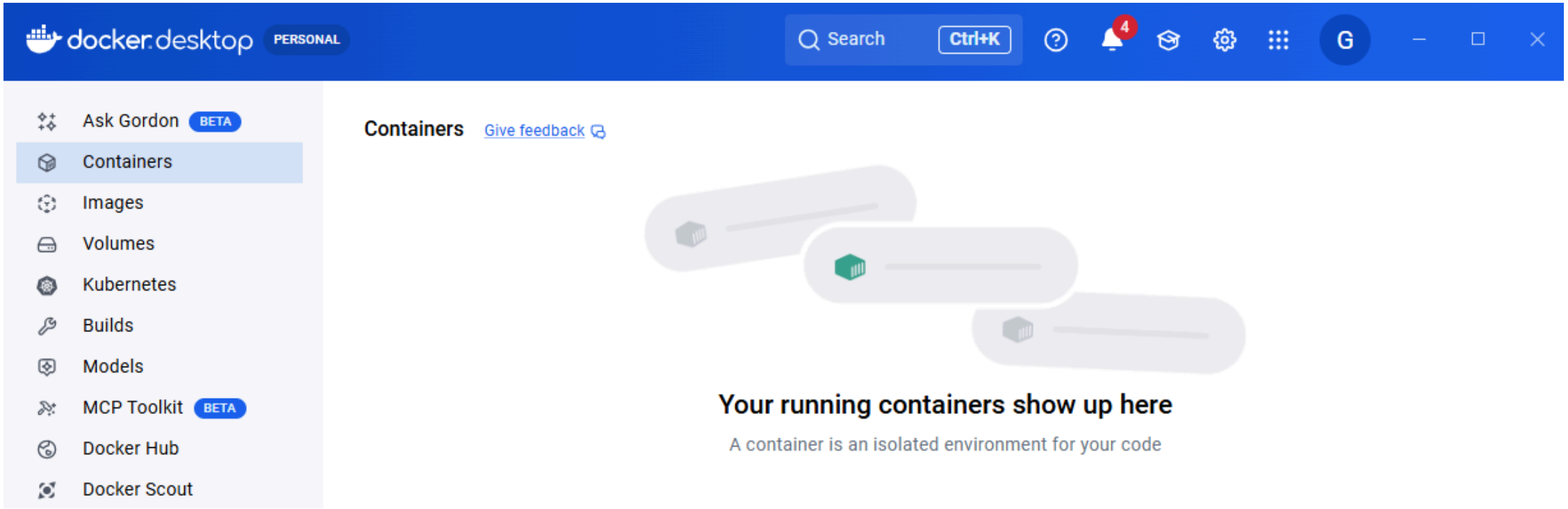
training_args.yaml

```yaml
train.warmup_steps: 0

# LoRA 구성
train.lora_rank: 16
train.lora_alpha: 32
train.lora_dropout: 0.05
train.lora_target: ""
train.use_dora: true

# 출력 디렉토리 (saves/{model}/{finetuning_type}/ 아래 폴더명)
train.output_dir: train_illnesses_dora
```

Docker

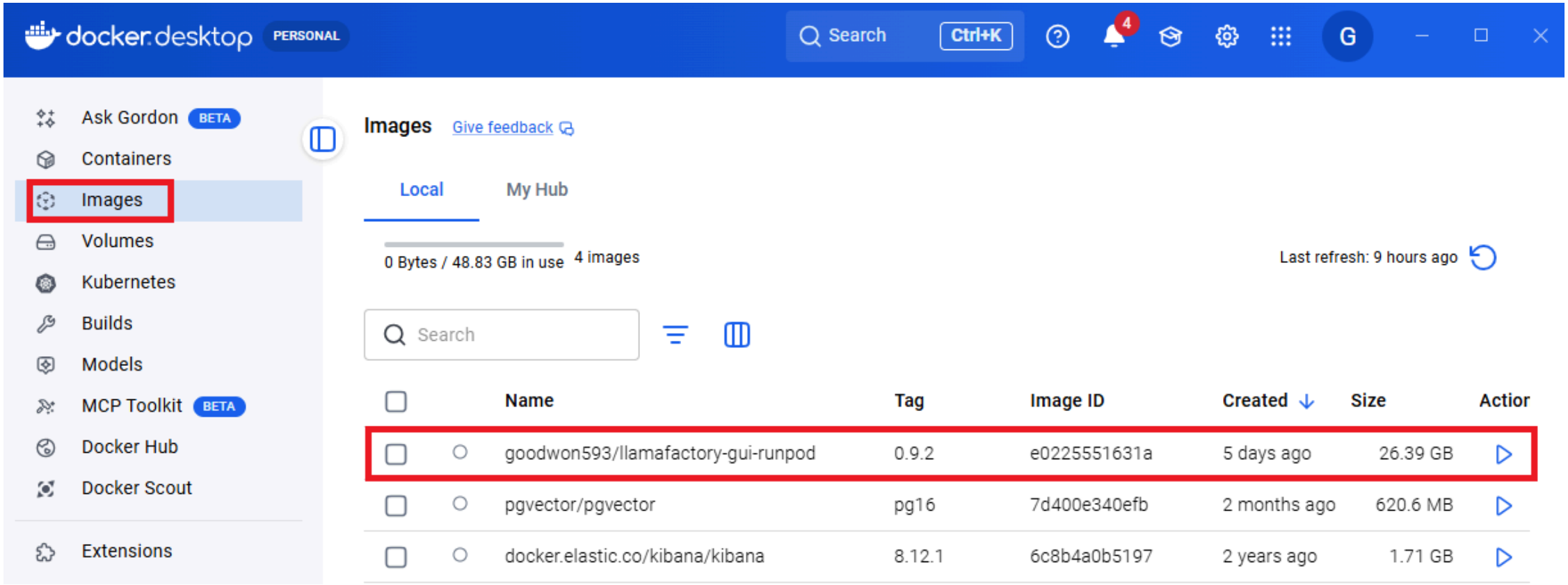

docker desktop
PERSONAL
Search
Ctrl+K
4
G
Ask Gordon
BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Docker Scout
Containers
Give feedback
Your running containers show up here
A container is an isolated environment for your code

단계1: Docker 이미지 빌드

```
docker build --platform linux/amd64 -t [YOUR_USERNAME]/llamafactory-gui-runpod:0.9.2 .
```

```
Problems   Output   Debug Console   Terminal   Ports                                                    powershell

PS C:\dev\course\course_LLM\2. LLM Fine Tuning\2. Fine Tuning with Runpod\3. 모델 학습\2-1. Llama Factory - GUI> docker build --platform linux/amd64 -t goodwon593/llamafactory-gui-r
unpod:0.9.2 .
[+] Building 2.4s (13/13) FINISHED                                                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                                                    0.0s
 => => transferring dockerfile: 1.43kB                                                                                                                                  0.0s
 => [internal] load metadata for docker.io/hiyouga/llamafactory:0.9.4                                                                                                  2.0s
 => [auth] hiyouga/llamafactory:pull token for registry-1.docker.io                                                                                                     0.0s
 => [internal] load .dockerignore                                                                                                                                       0.0s
 => => transferring context: 2B                                                                                                                                         0.0s
 => [1/7] FROM docker.io/hiyouga/llamafactory:0.9.4@sha256:ccf6329c69038a571059675651e7da2a3b85fdc7ded8b2ded93413dbe734cf7f                                              0.0s
 => => resolve docker.io/hiyouga/llamafactory:0.9.4@sha256:ccf6329c69038a571059675651e7da2a3b85fdc7ded8b2ded93413dbe734cf7f                                              0.0s
 => [internal] load build context                                                                                                                                       0.0s
 => => transferring context: 102B                                                                                                                                       0.0s
```


docker desktop
PERSONAL
Search
Ctrl+K
4
G
Ask Gordon
BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Docker Scout
Extensions
Images
Give feedback
Local
My Hub
0 Bytes / 48.83 GB in use
4 images
Last refresh: 9 hours ago
Search
Name
Tag
Image ID
Created
Size
Actior
goodwon593/llamafactory-gui-runpod
0.9.2
e0225551631a
5 days ago
26.39 GB
pgvector/pgvector
pg16
7d400e340efb
2 months ago
620.6 MB
docker.elastic.co/kibana/kibana
8.12.1
6c8b4a0b5197
2 years ago
1.71 GB

단계2: Docker Hub 배포

```
docker push [YOUR_USERNAME]/llamafactory-gui-runpod:0.9.2
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\dev\course\course_LLM\2. LLM Fine Tuning\2. Fine Tuning with Runpod\3. 모델 학습\2-1. Llama Factory - GUI> docker push goodwon593/llamafactory-gui-runpod:0.9.2
The push refers to repository [docker.io/goodwon593/llamafactory-gui-runpod]
e0383d3d5655: Waiting
c39c6f3a22d8: Waiting
68ffe0f88283: Waiting
ec6d5f6c9ed9: Waiting
47b8539d532f: Waiting
e28e706b01bf: Waiting
56dc85502937: Waiting
3e14b472177a: Waiting
7021d1b70935: Waiting
```

hub Explore My Hub

Search Docker Hub CtrlK

G

goodwon593
Docker Personal

- Repositories
- Hardened Images
- Collaborations
- Settings
  - Default privacy
  - Notifications
- Billing
- Usage
  - Pulls
  - Storage

Repositories / llamafactory-gui-runpod / General

goodwon593/llamafactory-gui-runpod

Last pushed 2 minutes ago · Repository size: 31.7 GB · 0 · 120

Add a description

Add a category

General Tags Image Management Collaborators Webhooks Settings

Tags

DOCKER SCOUT INACTIVE
Activate

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
0.9.2		Image	less than 1 day	2 minutes

Docker commands

Public view

To push a new tag to this repository:

```
docker push goodwon593/llamafactory-gui-runpod:tagname
```

buildcloud

Build with
Docker Build Cloud

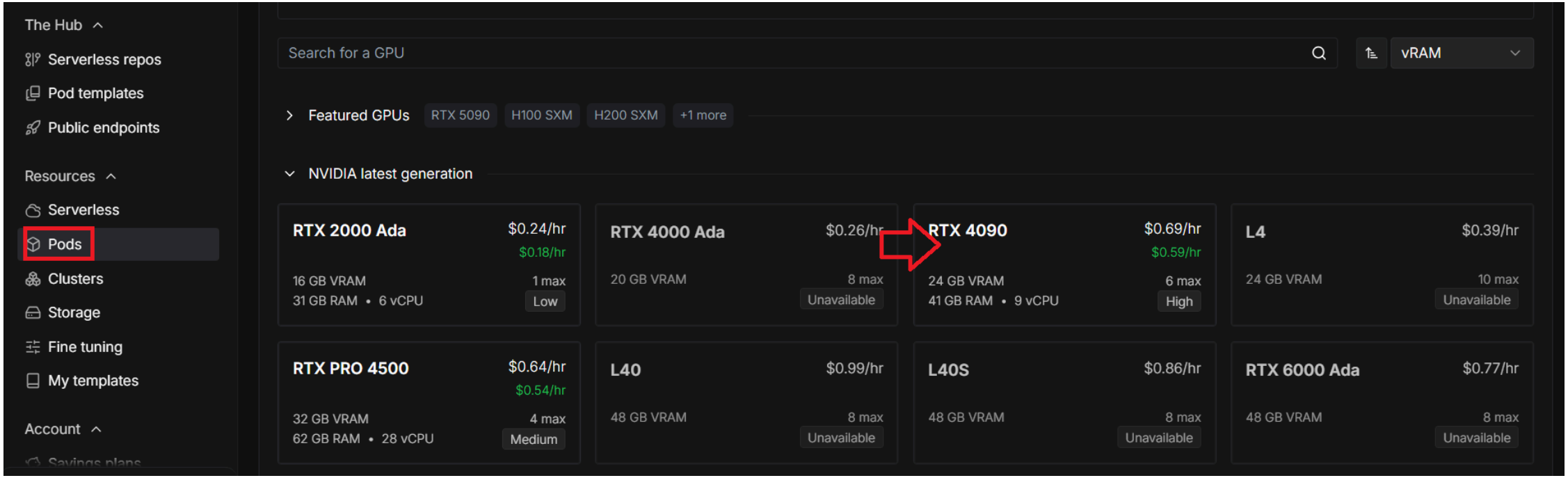
Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation

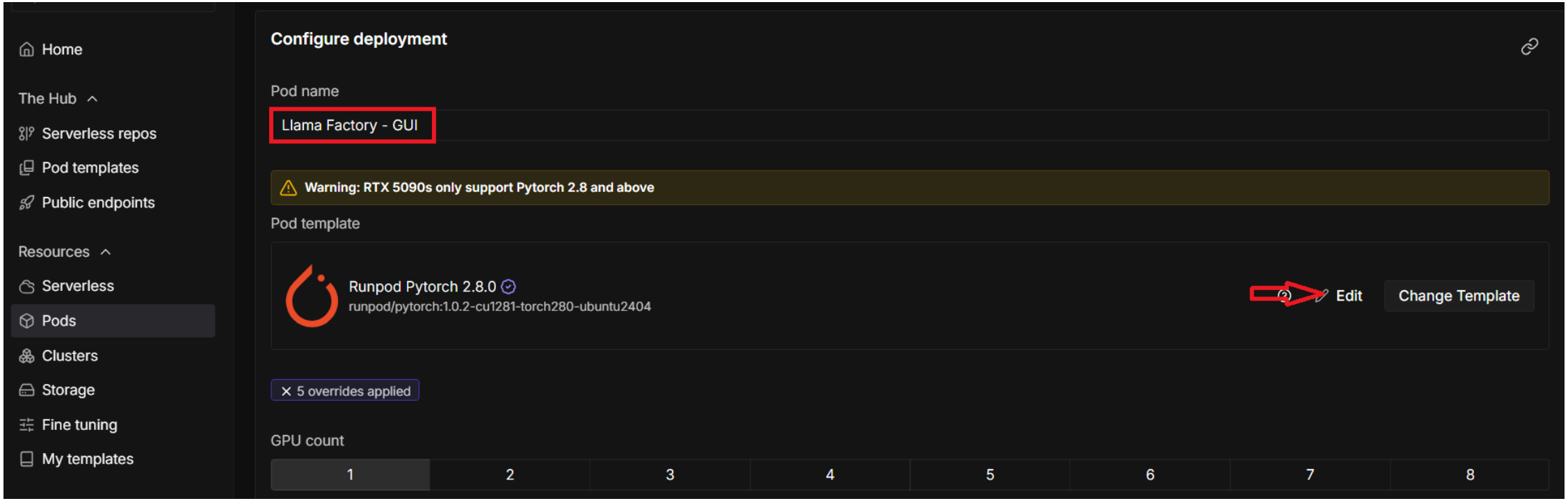
Runpod 모델 학습

단계1: GPU Pod 생성

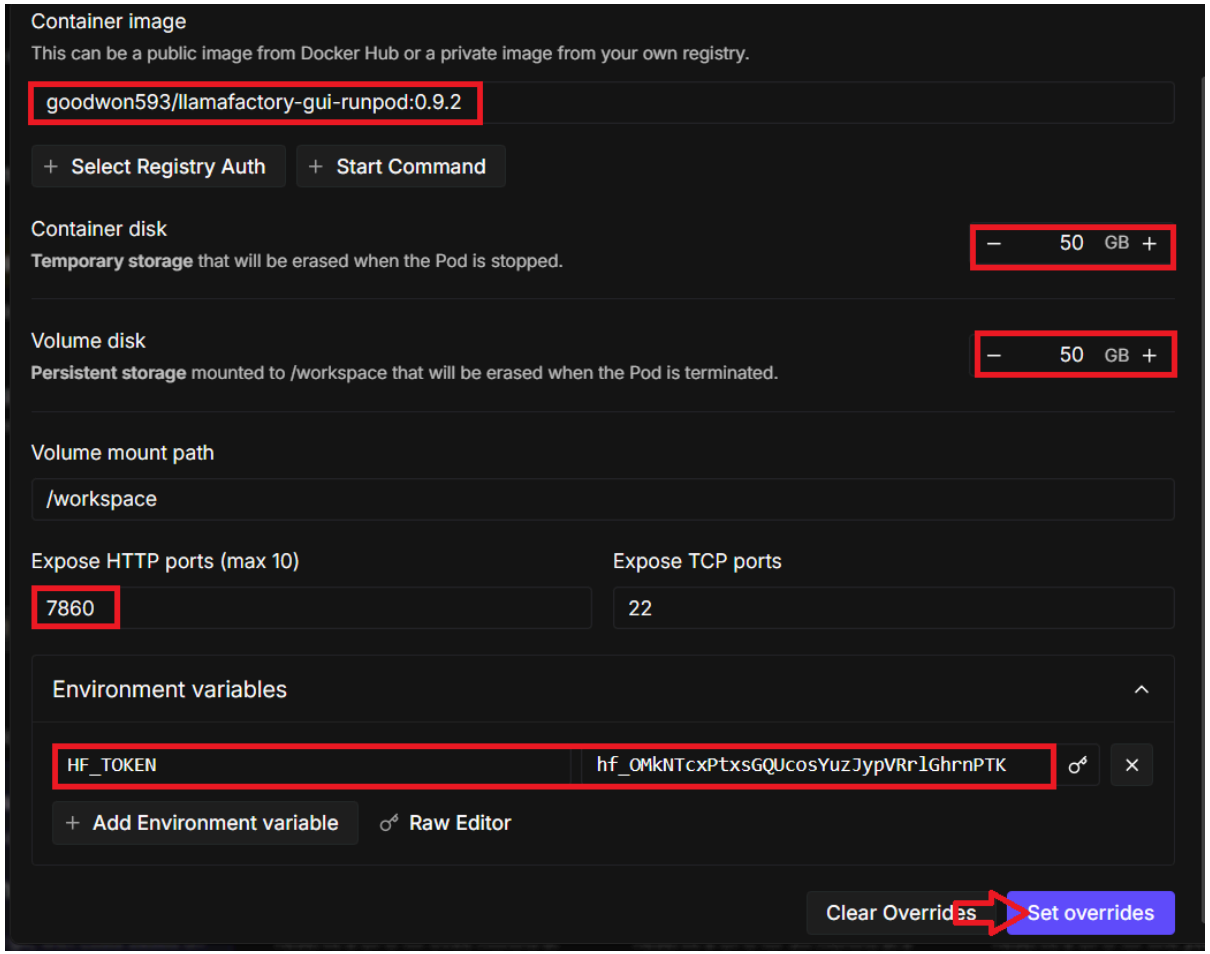
- RTX 5090 → Blackwell(sm_120) → 0.9.4 이미지와 호환 안됨
- 그 외 모든 GPU(RTX 3090, RTX 4090 (24GB)) → Ampere/Ada 계열 → 0.9.4 이미지로 DoRA 포함 정상 작동



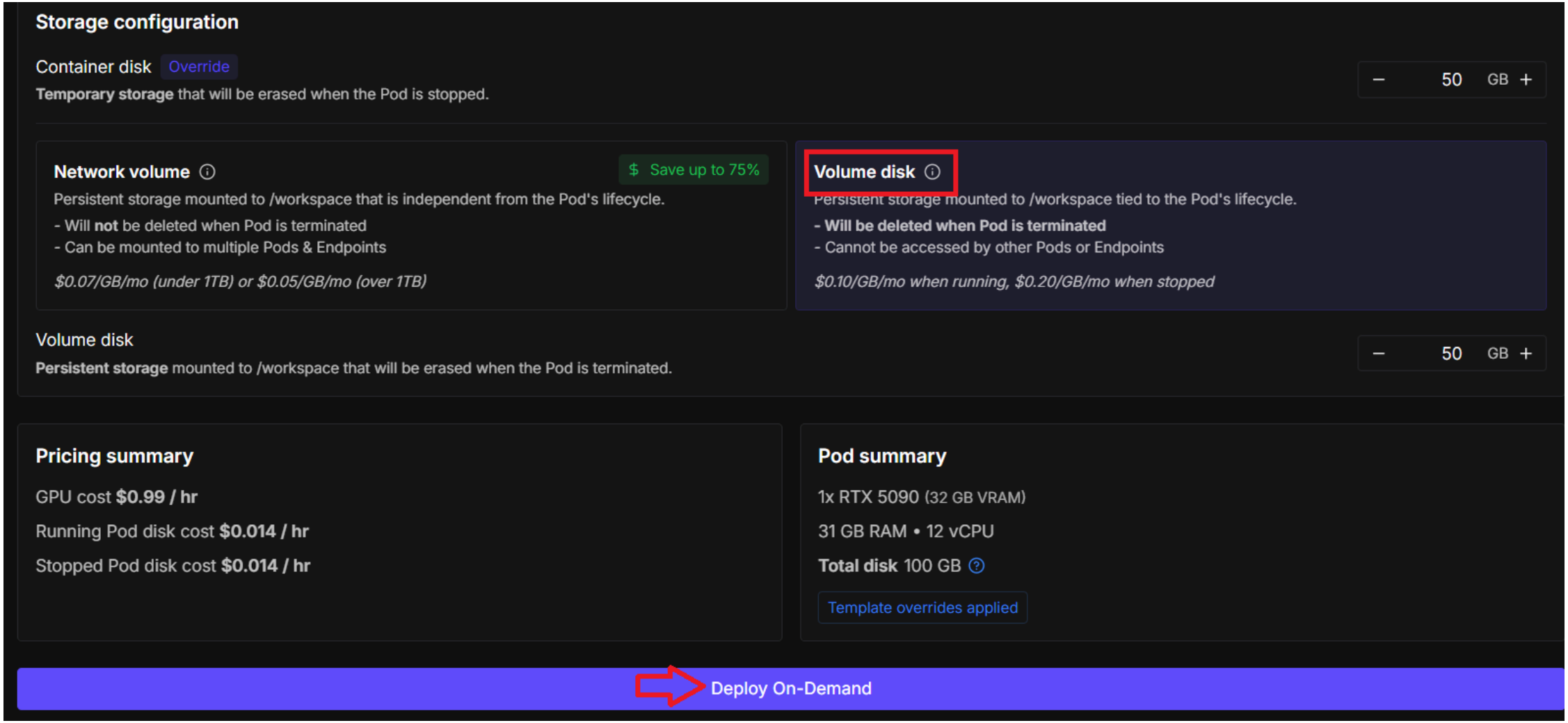
Configure deployment



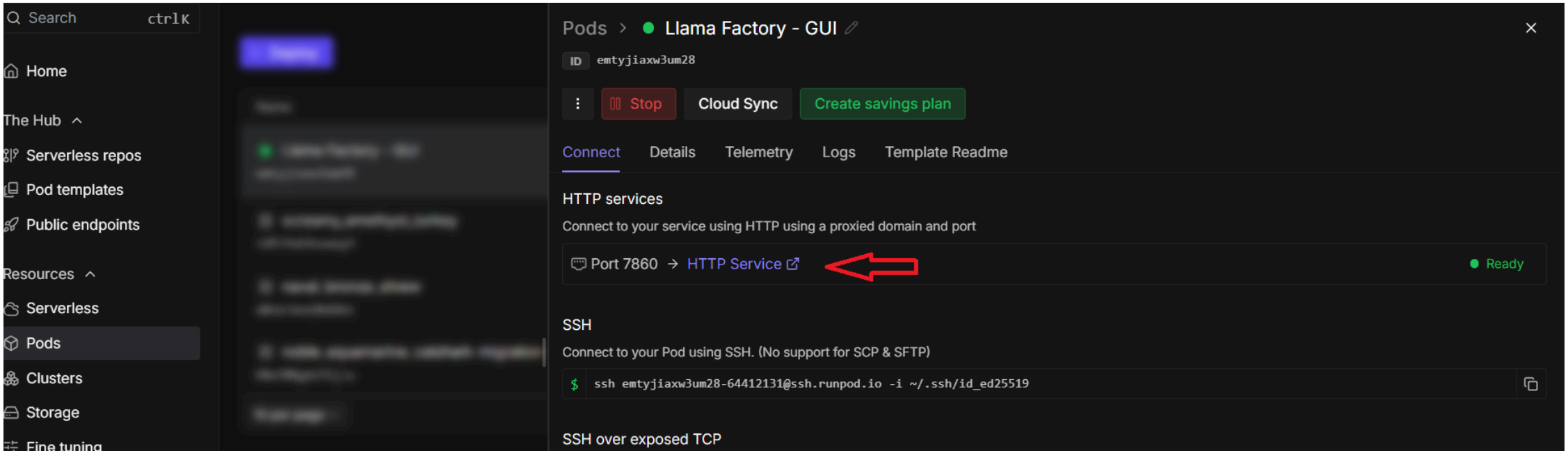
Pod template overrides



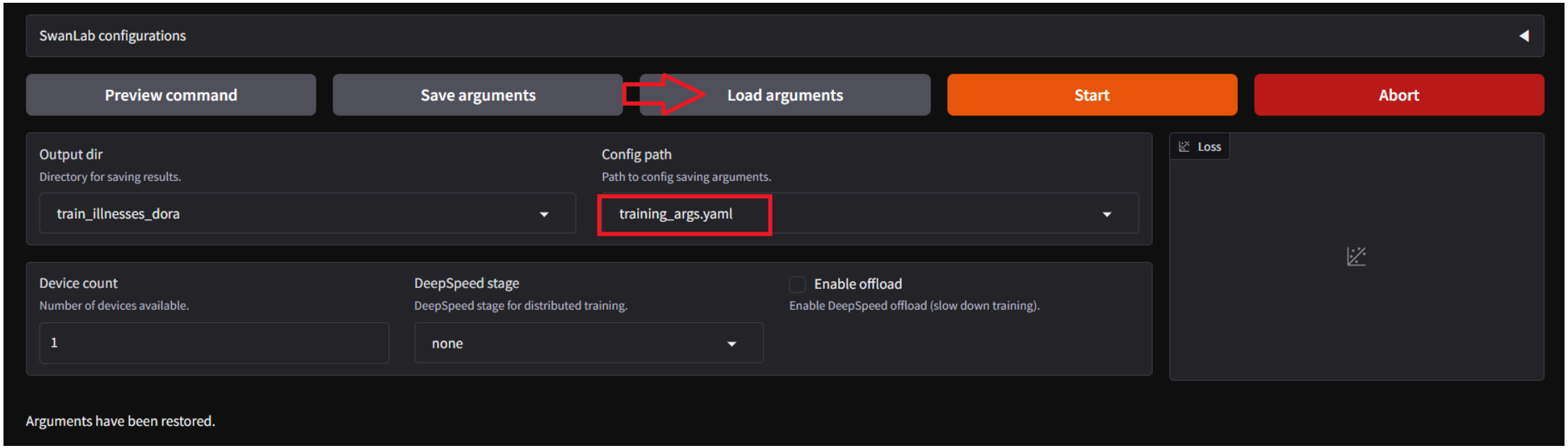
Deploy On-Demand

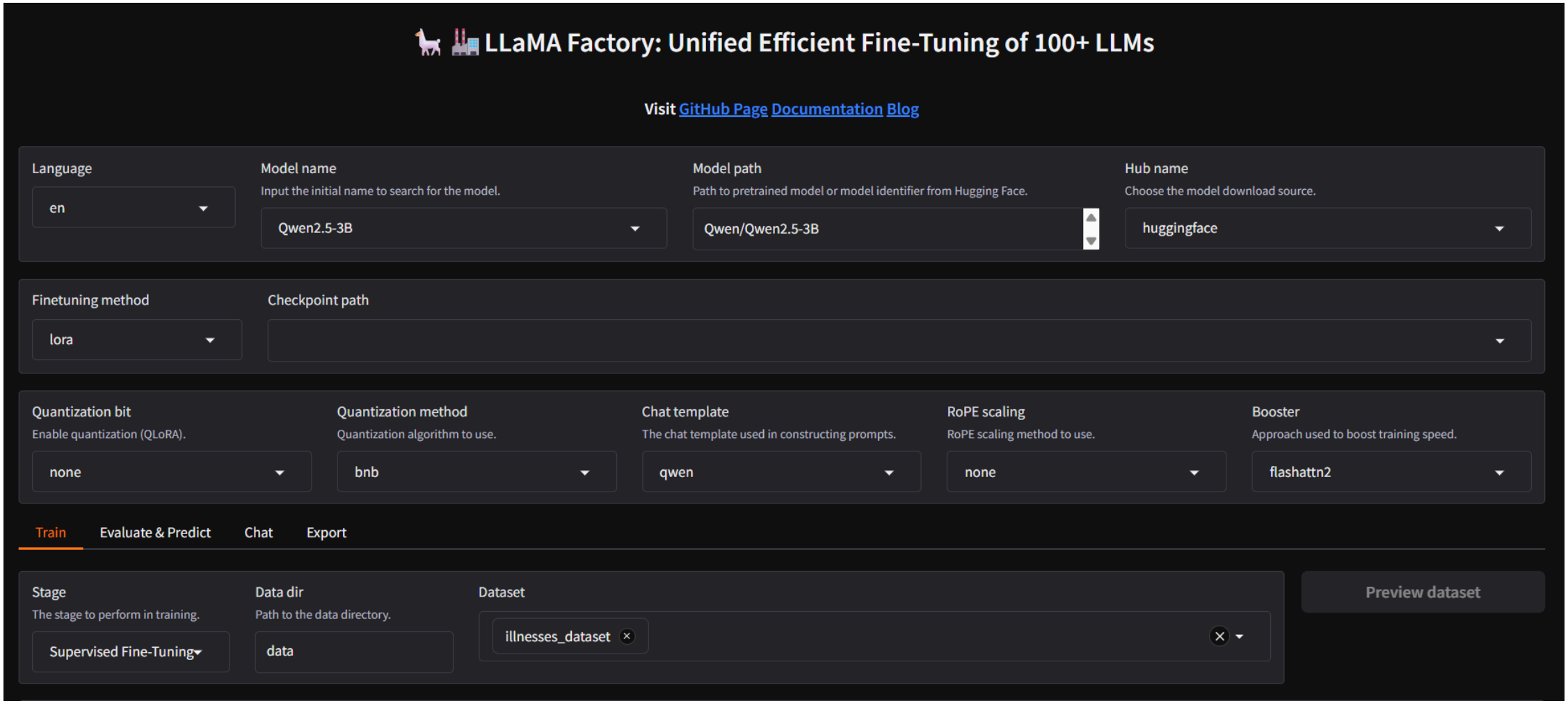


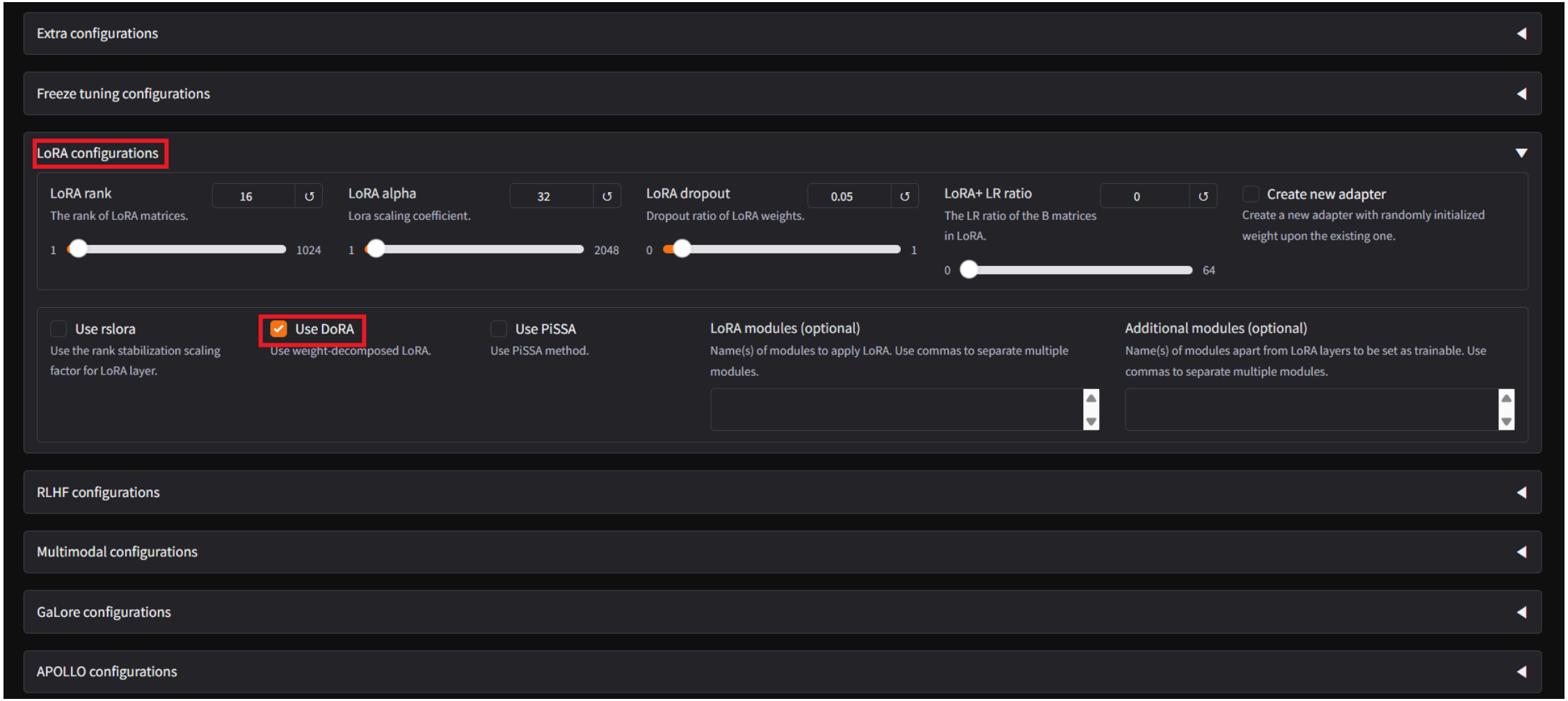
단계2: Llama Factory WebUI 접속



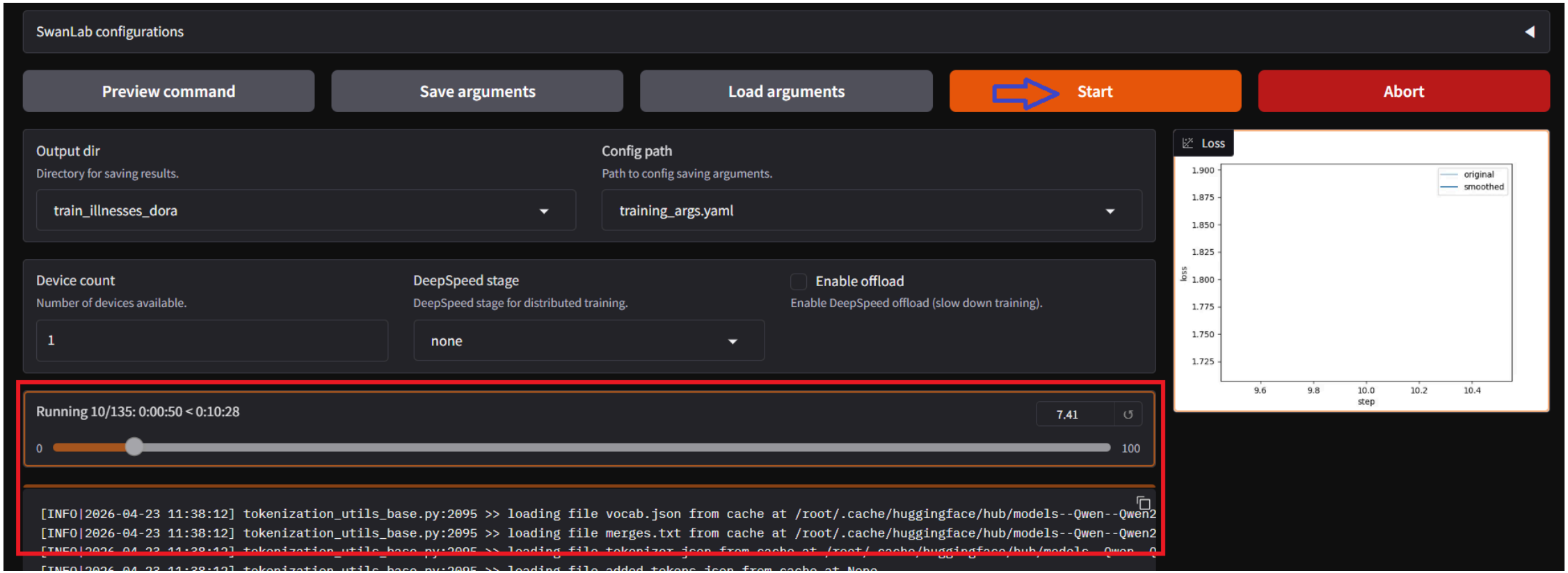
단계3: training_args.yaml 적용



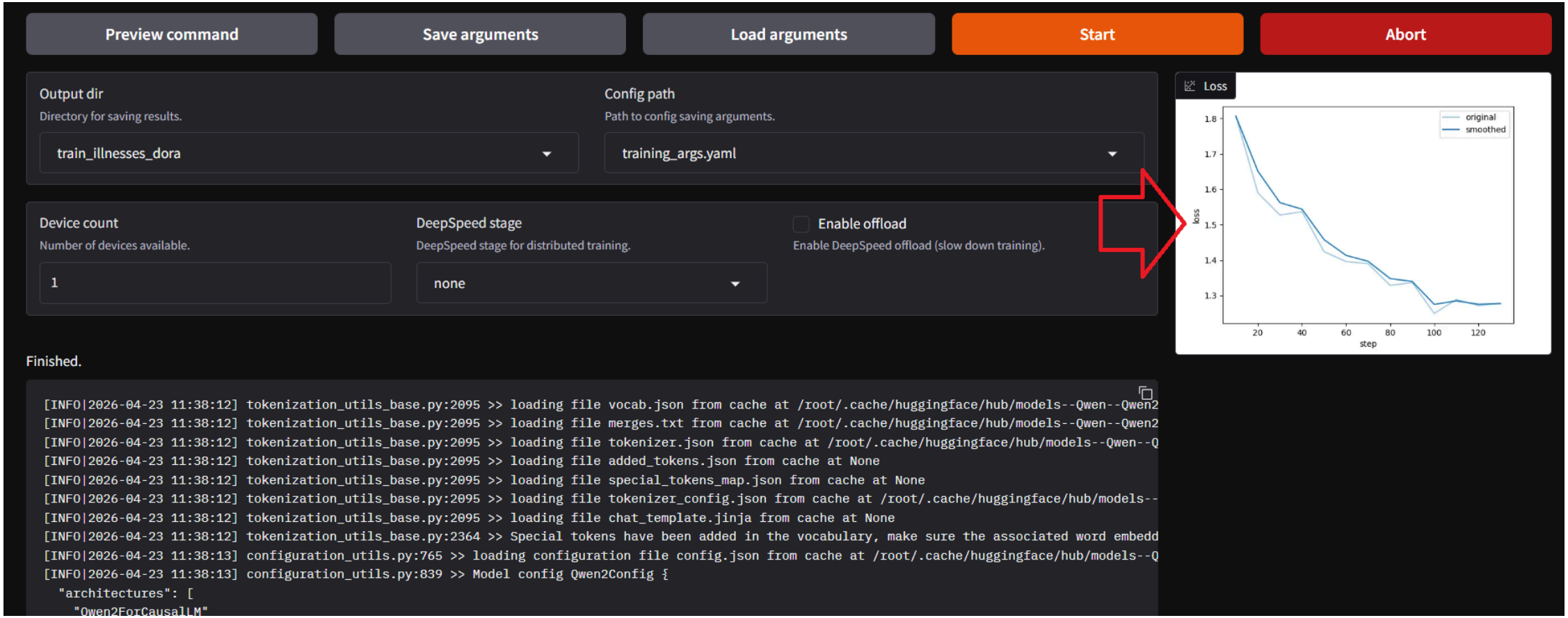

🦙🏭 LLaMA Factory: Unified Efficient Fine-Tuning of 100+ LLMs
Visit GitHub Page Documentation Blog
Language
en
Model name
Input the initial name to search for the model.
Qwen2.5-3B
Model path
Path to pretrained model or model identifier from Hugging Face.
Qwen/Qwen2.5-3B
Hub name
Choose the model download source.
huggingface
Finetuning method
lora
Checkpoint path
Quantization bit
Enable quantization (QLoRA).
none
Quantization method
Quantization algorithm to use.
bnb
Chat template
The chat template used in constructing prompts.
qwen
RoPE scaling
RoPE scaling method to use.
none
Booster
Approach used to boost training speed.
flashattn2
Train
Evaluate & Predict
Chat
Export
Stage
The stage to perform in training.
Supervised Fine-Tuning
Data dir
Path to the data directory.
data
Dataset
illnesses_dataset
Preview dataset


Extra configurations
Freeze tuning configurations
LoRA configurations
LoRA rank
16
The rank of LoRA matrices.
1
1024
LoRA alpha
32
Lora scaling coefficient.
1
2048
LoRA dropout
0.05
Dropout ratio of LoRA weights.
0
1
LoRA+ LR ratio
0
The LR ratio of the B matrices in LoRA.
0
64
Create new adapter
Create a new adapter with randomly initialized weight upon the existing one.
Use rslora
Use the rank stabilization scaling factor for LoRA layer.
Use DoRA
Use weight-decomposed LoRA.
Use PiSSA
Use PiSSA method.
LoRA modules (optional)
Name(s) of modules to apply LoRA. Use commas to separate multiple modules.
Additional modules (optional)
Name(s) of modules apart from LoRA layers to be set as trainable. Use commas to separate multiple modules.
RLHF configurations
Multimodal configurations
GaLore configurations
APOLLO configurations

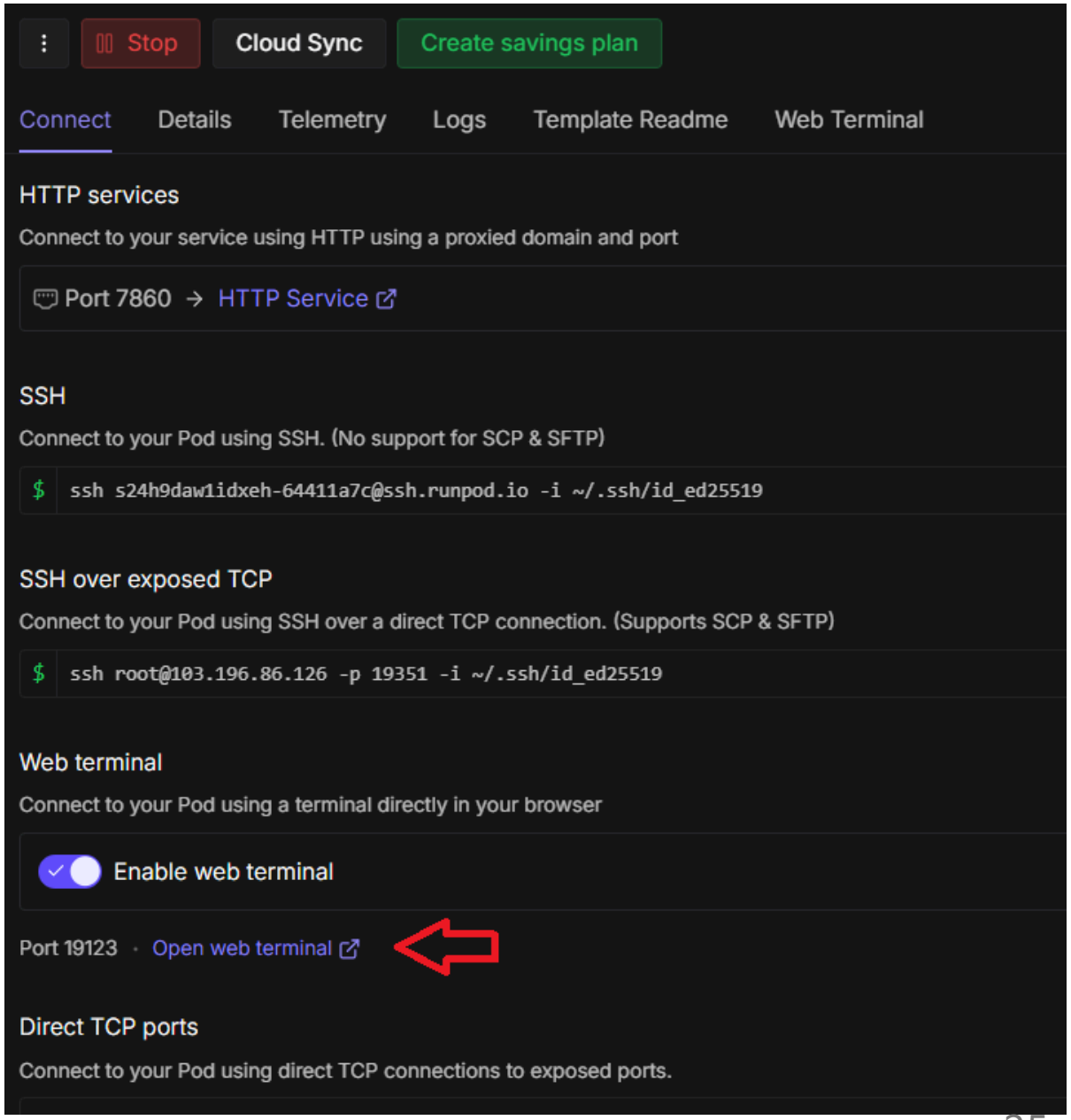
단계4: 훈련 시작


SwanLab configurations
Preview command
Save arguments
Load arguments
Start
Abort
Output dir
Directory for saving results.
train_illnesses_dora
Config path
Path to config saving arguments.
training_args.yaml
Device count
Number of devices available.
1
DeepSpeed stage
DeepSpeed stage for distributed training.
none
Enable offload
Enable DeepSpeed offload (slow down training).
Running 10/135: 0:00:50 < 0:10:28
7.41
0
100
[INFO|2026-04-23 11:38:12] tokenization_utils_base.py:2095 >> loading file vocab.json from cache at /root/.cache/huggingface/hub/models--Qwen--Qwen2
[INFO|2026-04-23 11:38:12] tokenization_utils_base.py:2095 >> loading file merges.txt from cache at /root/.cache/huggingface/hub/models--Qwen--Qwen2
Loss
original
smoothed
loss
step

학습 완료

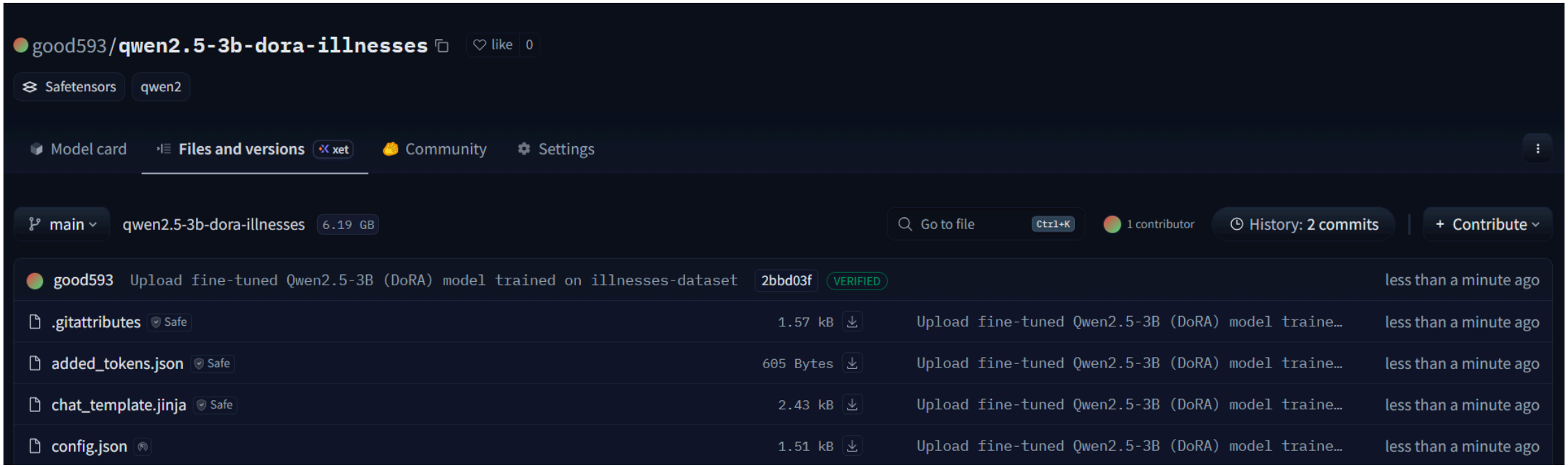

Preview command
Save arguments
Load arguments
Start
Abort
Output dir
Directory for saving results.
train_illnesses_dora
Config path
Path to config saving arguments.
training_args.yaml
Device count
Number of devices available.
1
DeepSpeed stage
DeepSpeed stage for distributed training.
none
Enable offload
Enable DeepSpeed offload (slow down training).
Loss
original
smoothed
loss
step
Finished.
[INFO|2026-04-23 11:38:12] tokenization_utils_base.py:2095 >> loading file vocab.json from cache at /root/.cache/huggingface/hub/models--Qwen--Qwen2
[INFO|2026-04-23 11:38:12] tokenization_utils_base.py:2095 >> loading file merges.txt from cache at /root/.cache/huggingface/hub/models--Qwen--Qwen2
[INFO|2026-04-23 11:38:12] tokenization_utils_base.py:2095 >> loading file tokenizer.json from cache at /root/.cache/huggingface/hub/models--Qwen--Q
[INFO|2026-04-23 11:38:12] tokenization_utils_base.py:2095 >> loading file added_tokens.json from cache at None
[INFO|2026-04-23 11:38:12] tokenization_utils_base.py:2095 >> loading file special_tokens_map.json from cache at None
[INFO|2026-04-23 11:38:12] tokenization_utils_base.py:2095 >> loading file tokenizer_config.json from cache at /root/.cache/huggingface/hub/models--
[INFO|2026-04-23 11:38:12] tokenization_utils_base.py:2095 >> loading file chat_template.jinja from cache at None
[INFO|2026-04-23 11:38:12] tokenization_utils_base.py:2364 >> Special tokens have been added in the vocabulary, make sure the associated word embedd
[INFO|2026-04-23 11:38:13] configuration_utils.py:765 >> loading configuration file config.json from cache at /root/.cache/huggingface/hub/models--Q
[INFO|2026-04-23 11:38:13] configuration_utils.py:839 >> Model config Qwen2Config {
"architectures": [
"Qwen2ForCausalLM"

단계5: HuggingFace 업로드



Upload model to HuggingFace Hub

```
bash /app/upload_model.sh
```

```
root@f9b6ccf05386:/app/LLaMA-Factory# bash /app/upload_model.sh
==================================================
  HuggingFace 모델 업로드 시작
  베이스 모델 : Qwen/Qwen2.5-3B
  어댑터 경로 : /app/LLaMA-Factory/saves/Qwen2.5-3B/lora/train_illnesses_dora
  업로드 대상 : good593/qwen2.5-3b-dora-illnesses
==================================================
[INFO] 어댑터 디렉토리 확인 완료: /app/LLaMA-Factory/saves/Qwen2.5-3B/lora/train_illnesses_do
ra
total 134M
-rw-r--r-- 1 root root 1.4K Apr 24 02:09 README.md
-rw-r--r-- 1 root root  926 Apr 24 02:09 adapter_config.json
```

HuggingFace Hub에서 업로드된 모델 확인